Robust Common Spatial Patterns based on Bhattacharyya Distance and Gamma Divergence

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***Abstract*— The computation of task-related spatial filters is a prerequisite for a successful application of motor-imagery based Brain-Computer Interfaces. However, in the presence of artifacts, e.g., resulting from eye movements or muscular activity, standard methods such as Common Spatial Patterns (CSP) perform poorly. Recently, a divergence-based spatial filter computation framework has been proposed which allows one to significantly robustify the computation against artifacts by using Beta divergence. In this paper we integrate two additional divergence measures, namely Bhattacharyya distance and Gamma divergence, into the divergence-based CSP framework and evaluate their robustness using simulations and the data set IVa from the BCI Competition III.**

I. Introduction

Brain-Computer Interfacing (BCI) [1] [2] serves as a non-muscular communication system, which detects brain signals from the electroencephalogram (EEG) and translates them into control commands for a computer device. People affected by diseases such as amyotrophic lateral sclerosis (ALS), brainstem stroke, multiple sclerosis or muscular dystrophies, and especially those who are completely locked-in could benefit from this technology since it provides a way to communicate without any muscular control. A popular mental strategy is so-called motor imagery. It provides a system based on imagined movements that cause significant and detectable changes in EEG. The main challenge in constructing a BCI device is extracting relevant features from a high-dimensional EEG to translate brain signals efficiently into control signals [3], [4], [5].

Common Spatial Patterns (CSP) is a well established feature-extraction algorithm in motor imagery based BCIs [6] [7]. It detects synchronization and desynchronization processes and uses them to compute spatial filters which make it possible to discriminate between different imagined movements. Since original CSP is sensitive to nonstationarities different robust CSP versions have been proposed over the last few years [8] [9] [10] [11]. Thinking about end users, BCI efficiency still has to be enhanced such that it can work in an out-of-lab context. One important issue is the robustness of the system. It has recently been shown that CSP can be embedded in a divergence framework using Beta divergence which is robust to outliers [12][13]. Divergence functions [14] are popular measures of discrepancy between probability distributions. Integrating them into machine learning methods is not entirely new, as different algorithms already have been transformed using divergence functions [15] [16] [17]. In this paper, we integrate Bhattacharyya distance and Gamma divergence into the divergence-based CSP (divCSP) framework and evaluate their robustness.

Detailed derivations and an implementation of the proposed algorithms are available at xxxxx.

II. Divergence-based framework for CSP

Spatial filtering is a common way to detect discriminative features by enhancing the signal-to-noise ratio in motor imagery. A well-established spatial filter algorithm is Common Spatial Patterns (CSP). Motor imagery usually comes along with synchronization and desynchronization effects (ERS/ERD) in mu and beta rhythms over the sensorimotor cortex during and after imagined movements [18]. CSP detects those effects by maximizing the variance of class one while minimizing the variance of class two and vice versa. This problem can be solved by a generalized eigenvalue problem

$$\Sigma_1 w_i = \lambda_1 \Sigma_2 w_i$$

where Σ_1 and $\Sigma_2 \in \mathbb{R}^{\mathrm{DxD}}$ are the average covariance matrices of class 1 and 2, w_i are the spatial filters and λ_i the corresponding eigenvalues. The obtained spatial filters $W = [w_1, w_2, \ldots, w_D]$ then have to be sorted according to their contributing discriminative quality [13]. In [12] it was shown that the subspace spanned by those spatial filters is equal to the span of filters which maximize the symmetric Kullback-Leibler (KL) divergence between the probability distribution of both classes

$$\mathrm{span}(W) = \mathrm{span}(V^*)$$
$$V^* = \underset{V}{\mathrm{argmax}}\, \tilde{D}_{kl}\left(\mathcal{N}\left(0, V^\top \Sigma_1 V\right) \,||\, \mathcal{N}\left(0, V^\top \Sigma_2 V\right)\right)$$

where V^* maximizes the symmetric KL-divergence between the probability distribution of both classes and $\mathcal{N}(\mu, \Sigma)$ denotes the Gaussian distribution with mean μ and covariance Σ. Symmetric KL-divergence between two continuous probability distributions $p(x)$ and $q(x)$ is defined as:

$$\tilde{D}_{kl} = \int p(x) \log \frac{p(x)}{q(x)} dx + \int q(x) \log \frac{q(x)}{p(x)} dx$$

Since KL-divergence is sensitive to outliers, we investigate robust divergence measures to enhance classification accuracy.

III. Robustness and robust discrepancy measures

A. Robustness Property

The goal of a robust CSP algorithm is to reliably compute task-related spatial filters V even when data is heavily contaminated. In the divergence framework robustness can be achieved by decomposing the divergence between the average class distributions into the sum of trialwise divergences

TABLE I: Overview over objective function and properties of the different CSP divergence methods with $\bar{\Sigma}_c^i = V^T \Sigma_c^i V$ and $\bar{\Sigma}_c$ denoting the projected covariance matrix of trial i and class c and the class average. Note that $\rho = \frac{1}{\beta}\sqrt{\frac{1}{(2\pi)^{\beta d}(\beta+1)^d}}$.

Method	**Objective** $\sigma(V)$	**robust**	**parameter**
CSP	$\frac{1}{2}\operatorname{tr}((\bar{\Sigma}_1)^{-1}(\bar{\Sigma}_2)) + \frac{1}{2}\operatorname{tr}((\bar{\Sigma}_2)^{-1}(\bar{\Sigma}_1)) - d$	no	no
Beta	$\rho \sum_{i=1}^n (\lvert\bar{\Sigma}_1^i\rvert^{-\frac{\beta}{2}} + \lvert\bar{\Sigma}_2^i\rvert^{-\frac{\beta}{2}} - (\beta+1)^{\frac{d}{2}}(\lvert\bar{\Sigma}_2^i\rvert^{\frac{1-\beta}{2}}\lvert\beta\bar{\Sigma}_1^i + \bar{\Sigma}_2^i\rvert^{-\frac{1}{2}} + \lvert\bar{\Sigma}_1^i\rvert^{\frac{1-\beta}{2}}\lvert\beta\bar{\Sigma}_2^i + \bar{\Sigma}_1^i\rvert^{-\frac{1}{2}}))$	yes	β
Bha	$\frac{1}{2}\sum_{i=1}^n (\ln(\lvert\bar{\Sigma}_1^i + \bar{\Sigma}_2^i\rvert) - \frac{1}{2}\ln\lvert\bar{\Sigma}_1^i\rvert - \frac{1}{2}\ln\lvert\bar{\Sigma}_2^i\rvert - d\ln(2))$	yes	no
Gamma	$\frac{1}{4\gamma}\sum_{i=1}^n (\frac{1}{2}\ln(\lvert\gamma\bar{\Sigma}_1^i + \bar{\Sigma}_2^i\rvert) + \frac{1}{2}\ln(\lvert\bar{\Sigma}_1^i + \gamma\bar{\Sigma}_2^i\rvert) - \ln\lvert\bar{\Sigma}_1^i\rvert - \ln\lvert\bar{\Sigma}_2^i\rvert - d\ln(2))$	yes	γ

and limiting the influence of single (potentially outlier) terms (see [12]). This changes the objective function into

$$V^* = \underset{V}{\operatorname{argmax}} \sum_i \tilde{D}_{kl}\left(\mathcal{N}\left(0, V^\top \Sigma_1^i V\right) \,||\, \mathcal{N}\left(0, V^\top \Sigma_2^i V\right)\right)$$

where Σ_c^i refers to trialwise covariance matrices of class c. Note that this assumes a balanced number of trials for each class. This approach also requires robust divergence measures. We call a divergence D *robust* if the inclusion of a single additional outlier trial (e.g. $\Sigma_1^{i^*}$) does not significantly increase the value of the objective function. Mathematically, this translates to stating that the ratio α between the objective functions (with and without the outlier trial) is close to 1, i.e.,

$$\alpha = \frac{\max\limits_V \left\{\sigma(V) + D\left(\mathcal{N}(0, V^\top \Sigma_1^{i^*} V) || \, \mathcal{N}(0, V^\top \Sigma_2^i V)\right)\right\}}{\max\limits_V \sigma(V)}$$

where $\sigma(V)$ is the corresponding objective function (see Table I). If this ratio is very large (e.g., because it grows linearly or exponentially with $||V^\top \Sigma_1^{i^*} V||$) then the outlier trial has a significant influence on the spatial filter computation. In other words the algorithm focuses on the outlier trial instead of considering the majority of the 'clean trials'.

B. Beta divergence

Beta divergence was considered as a robust replacement in the divergence CSP algorithm [12]. Beta divergence [19] between two continuous probability distributions is defined (for $\beta > 0$) as:

$$D_\beta(p(x) \,||\, q(x)) = \frac{1}{\beta}\int (q^\beta(x) - p^\beta(x))p(x)dx - \frac{1}{\beta+1}\int (q^{\beta+1}(x) - p^{\beta+1}(x))dx$$

The objective function of the Beta divergence CSP algorithm is defined in the second row of Table I where $\tilde{D}_\beta$ denotes the symmetric Beta divergence and Σ_1^i and Σ_2^i are trial-wise covariance matrices of classes 1 and 2, respectively. In the following we will show that Beta divergence is not the only well-suited robust distance measure in the divergence-based CSP framework.

C. Bhattacharyya distance

Bhattacharyya distance, a divergence-type measure between two populations [20], is widely used in various fields such as computer vision [21], multiclass classification [22] and bayesian classification [23]. The key advantages in using the Bhattacharyya distance in the divergence-based CSP framework are its easy evaluation and the lack of parameters which often makes calculations significantly more complicated. Bhattacharyya distance between two continuous probability distributions $p(x), q(x)$ is defined as:

$$D_{bha}(p(x) \,||\, q(x)) = -\ln\left(\int \sqrt{p(x)q(x)}dx\right)$$

Incorporating the Bhattacharyya distance into the divergence-based CSP framework of [13] leads to the objective function defined in the third row of Table I.

D. Gamma divergence

Gamma divergence, first defined in 2008 in [24], is considered as a *super robust*[1] divergence measure. It has been used in robust parameter estimation [24], robust blind source separation [25] and clustering algorithms [26]. Symmetric Gamma divergence between two continuous probability distributions $p(x) = p,\ q(x) = q$ is defined (for $\gamma > 0$) as:

$$\tilde{D}_\gamma(p \,||\, q) = \frac{1}{2}(d_\gamma(p,q) - d_\gamma(p,p) + d_\gamma(q,p) - d_\gamma(q,q))$$

where

$$d_\gamma(p,q) = -\frac{1}{\gamma}\log\left(\int p\, q^\gamma dx\right) + \frac{1}{1+\gamma}\log\left(\int q^{1+\gamma}dx\right)$$

The objective function of the Gamma divergence CSP algorithm is defined in the last row of Table I.

IV. Evaluation

A. Simulations

In the first simulation experiment we visualize the effect of a single outlier on the presented divergences. Therefore we chose a two dimensional example of covariance matrices where Σ_1 is constant over 100 trials and Σ_2 is constant over

[1]The phenomenal of that the breakdown point of an estimator is larger than 50% is termed super robustness.

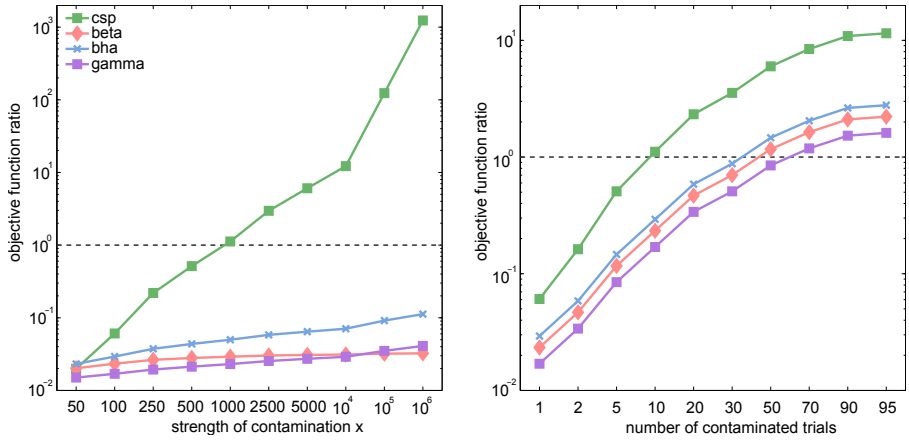

Fig. 1: Left: The impact of the single outlier on the objective function ratio. If the ratio increases 1 then the method will not extract the task-related source. Right: The objective function ratio after contaminating various amounts of trials.

99 trials and includes one outlier trial (Σ_2^{100})

$$\Sigma_1^{1:100} = \begin{pmatrix} 10 & 0 \\ 0 & 1 \end{pmatrix} \qquad \Sigma_2^{1:99} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \Sigma_2^{100} = \begin{pmatrix} 1 & 0 \\ 0 & x \end{pmatrix}$$

The correct discriminative source is the first one which would be detected by spatial filter $w_1 = (1,0)^T$. Depending on x a non robust method would be influenced by that outlier such that $w_2 = (0,1)^T$ would be preferred over w_1, i.e., the method would detect the wrong source. In the left panel of Figure 1 we visualize the ratio of the objective functions when applying the filters w_2 and w_1, i.e., $\frac{\sigma(w_2)}{\sigma(w_1)}$. One can see that the standard CSP algorithm is not robust because the ratio grows very fast and becomes larger than 1 (i.e., CSP quickly prefers w_2 over w_1). Beta divergence and the two discrepancy measures proposed in this paper successfully reduce the influence of the outlier and prefer the correct filter w_1 even for very large outlier values x.

In a second experiment we evaluate the robustness of the presented divergences when the number of outlier trials increases. We use the same example as before with an outlier value x of 100. We add an increasing number of outliers to the data and compute the ratio of the objective functions. The right panel of Figure 1 depicts the results. Also here we observe a significant increase in robustness when using the robust discrepancy measures. The standard CSP algorithm breaks down (ratio of objective functions exceeds 1) when more than 10 outliers are contained in the data whereas the Gamma divergence CSP method withstands up to 70 outlier trials in this example. Note that we have fixed the parameters β and γ to 0.2 in the presented simulation experiments.

B. BCI Competition Dataset

The described methods are applied to the data set IVa [27] from the BCI Competition III [28]. The data set contains EEG recordings from five healthy participants performing right hand and foot motor imagery without getting any feedback. The target class has been visualized by letters appearing behind a fixation cross and a randomly moving object. The EEG signal was recorded from 118 Ag/AgCl electrodes, band-pass filtered between 0.05 and 200 Hz and downsampled to 100 Hz. We manually select 68 electrodes mainly those covering the motor cortex and divide the data into training and testing set according to BCI Competition III. Parameters for gamma and beta divergence are determined individually for each participant by a 2-fold cross-validation using the following values: [0,0.0001,0.001,0.01,0.05,0.1,0.15,0.2,0.25,0.5,0.75,1,1.5,2,5] where 0 is equivalent to original CSP and $\gamma = 1$ is equivalent to Bha-divCSP.

Classification accuracies can be seen in Table II. It seems

TABLE II: Mean classification accuracies for different CSP methods

	aa	al	av	aw	ay
CSP	66.07	96.43	58.16	**88.84**	80.95
Beta	**74.11**	96.43	**70.41**	77.23	80.95
Bha	72.32	96.43	**70.41**	63.84	80.95
Gamma	**74.11**	96.43	69.39	83.04	80.95

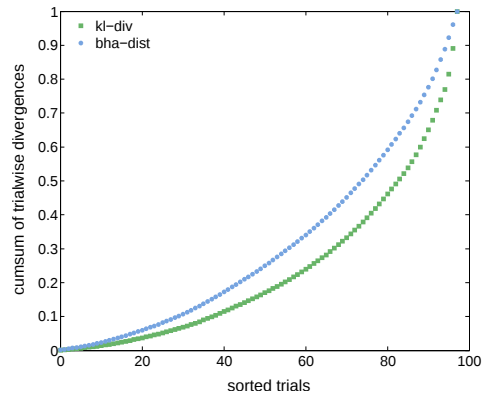

Fig. 2: Trialwise KL-divergences and Bhattacharyya distances of Bha-divCSP projected data of subject *av*. The divergences are sorted, normed and accumulated.

that especially the recordings of subjects *aa* and *av* are affected by artifacts because all robust divergence CSP methods improve classification accuracy over the standard CSP baseline. Due to the trade-off between robustness and efficiency not all subjects benefit from using robust divergence-based spatial filtering. Another effect which certainly also plays a role is the lack of data in the case of subjects *aw* and *ay*. Since these users only contain 56 (26 foot, 30 right) and 28 (10 foot, 18 right) training trials, respectively, where due to symmetry (see section III-A) even only 52 and 20 are used, they do not benefit from applying robust divergence-based CSP.

In Figure 2 we analyze the reason for the performance increase of subject *av*. For that we depict the impact of single trials on the sum in the objective function of the divergence CSP algorithms (see Table I). We assumed that for Bhattacharyya distance outlier trials with larger distance have smaller impact on the total sum than for KL divergence. This effect of downweighting becomes obvious by cumulating the sorted and normed trialwise distances. Bhattacharyya distance increases more linearly than KL divergence which means that those trials with largest distance have a major impact on the sum of KL divergences such that outliers are highly influencing the result.

V. Discussion

The ability to control a BCI in the real world represents a major challenge wherefore robustness plays an important role. Since brain recordings are highly sensitive and differ from brain to brain it is difficult to balance the trade-off between accuracy and robustness. Simulations and the application to real data have shown that the proposed discrepancy measures work significantly better than original CSP in case of heavy contamination. Selecting the optimal parameter highly complicate calculations especially when only few training data is available (e.g. participant *aw*). Future work could focus on a preselection regarding different csp methods and enhancing the parameter selection.

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